

E-Commerce Sales Data Exploratory Data Analysis

Customers • Products • Cities • Sales Trends • Correlations



PREPARED BY

Sandeep Kumar SankaraGuptam

INTERNSHIP

ApexPlanet Data Analytics Internship

DATE

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DATASET

728 records · 14 features · Structured CSV

1. Introduction & Objectives

Exploratory Data Analysis (EDA) summarizes a dataset's main characteristics using statistics and visualizations. This project analyzes an E-Commerce Sales dataset to uncover customer behaviour, product performance, sales trends, and revenue patterns — supporting decisions on pricing, inventory, and marketing.

- Understand dataset structure
 - Measure performance via KPIs
 - Analyze sales trends over time
 - Detect revenue outliers
- Analyze customer demographics
 - Identify top products / categories
 - Compare revenue across cities
 - Study variable relationships

DATASET OVERVIEW

Dataset	ApexPlanet E-Commerce Sales Dataset	Records	728
Features	14 (post feature engineering)	Domain	E-Commerce

Columns: Order ID, Order Date, Customer ID, Customer Name, Age, Gender, City, Product, Category, Quantity, Unit Price, Total Amount, Year, Month

DATA CLEANING & QUALITY

Missing Values	None	Duplicate Records	None
Invalid Dates	Corrected	Final Dataset	Ready for analysis

Steps: imported into Google Colab → verified data types, missing values & duplicates → converted Order Date to DateTime → corrected invalid dates → engineered Year/Month → exported the cleaned dataset for analysis.

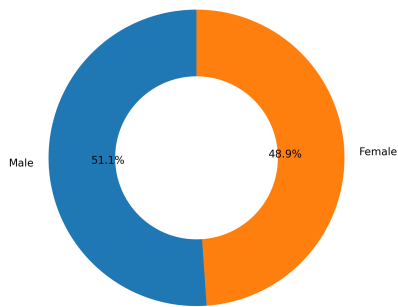
KPIs TRACKED

◆ Total Revenue	◆ Total Orders	◆ Total Customers
◆ Total Quantity Sold	◆ Avg. Order Value	◆ Avg. Unit Price

2. Customer & Product Analysis

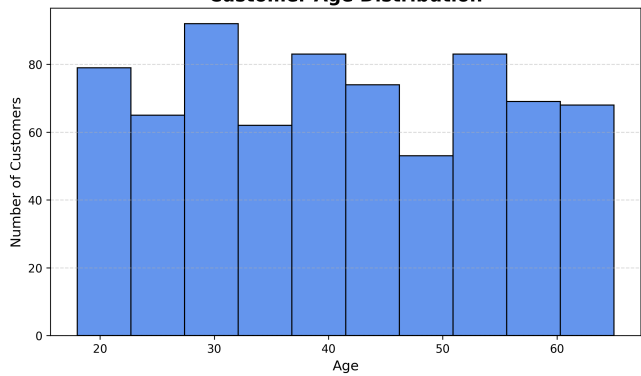
Understanding who buys, and what sells best.

Customer Gender Distribution



Customer Gender Distribution

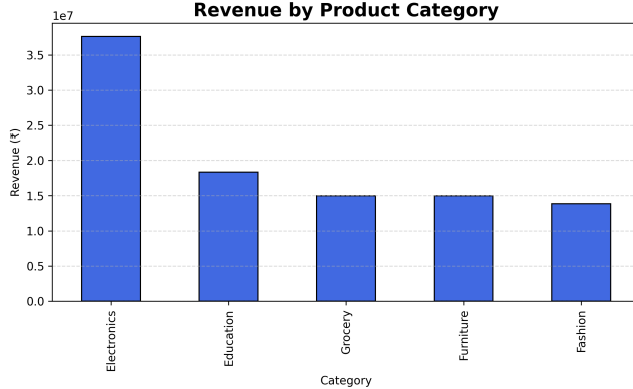
Customer Age Distribution



Customer Age Distribution

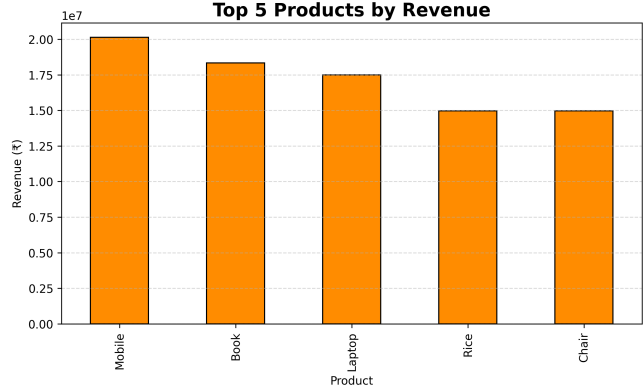
The customer base is nearly evenly split by gender (51% male / 49% female), and purchases are spread fairly evenly across age groups from 18–65, indicating a broad, non-age-restricted market.

Revenue by Product Category



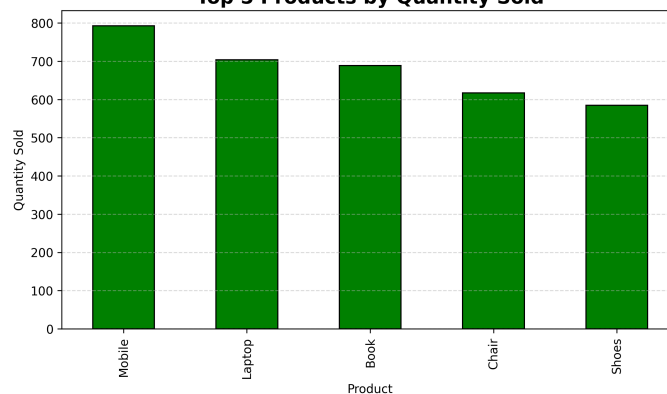
Revenue by Product Category

Top 5 Products by Revenue



Top 5 Products by Revenue

Top 5 Products by Quantity Sold

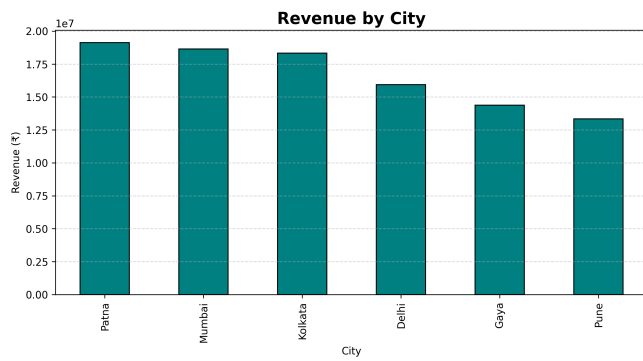


Top 5 Products by Quantity Sold

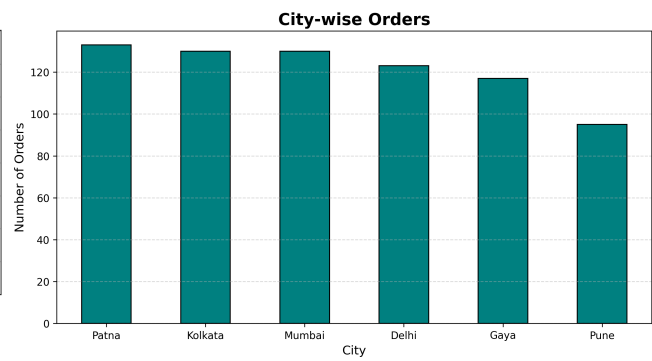
Electronics dominates category revenue, more than double the next category. Mobile, Book and Laptop lead individual product revenue, while Mobile also tops units sold — confirming it as the single strongest product line.

3. City & Time-Series Analysis

Where the business performs best, and how sales move through the year.

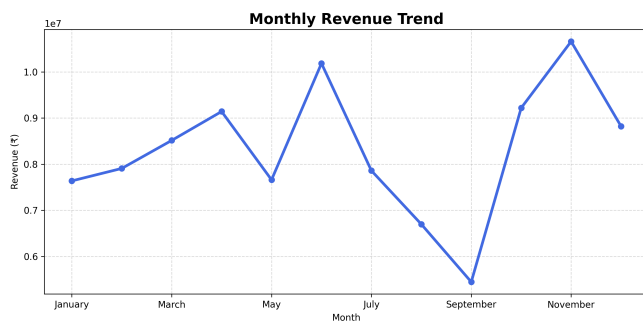


Revenue by City



Orders by City

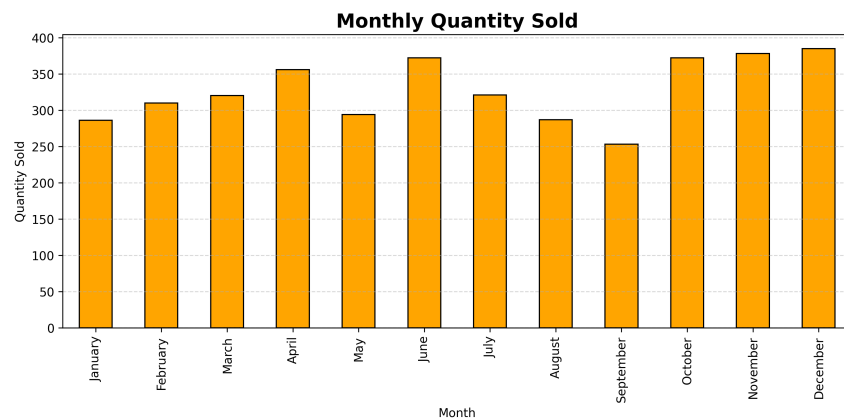
Patna, Mumbai and Kolkata are the top revenue and order-generating cities, while Pune trails the group — a candidate for focused regional marketing.



Monthly Revenue Trend



Monthly Orders Trend

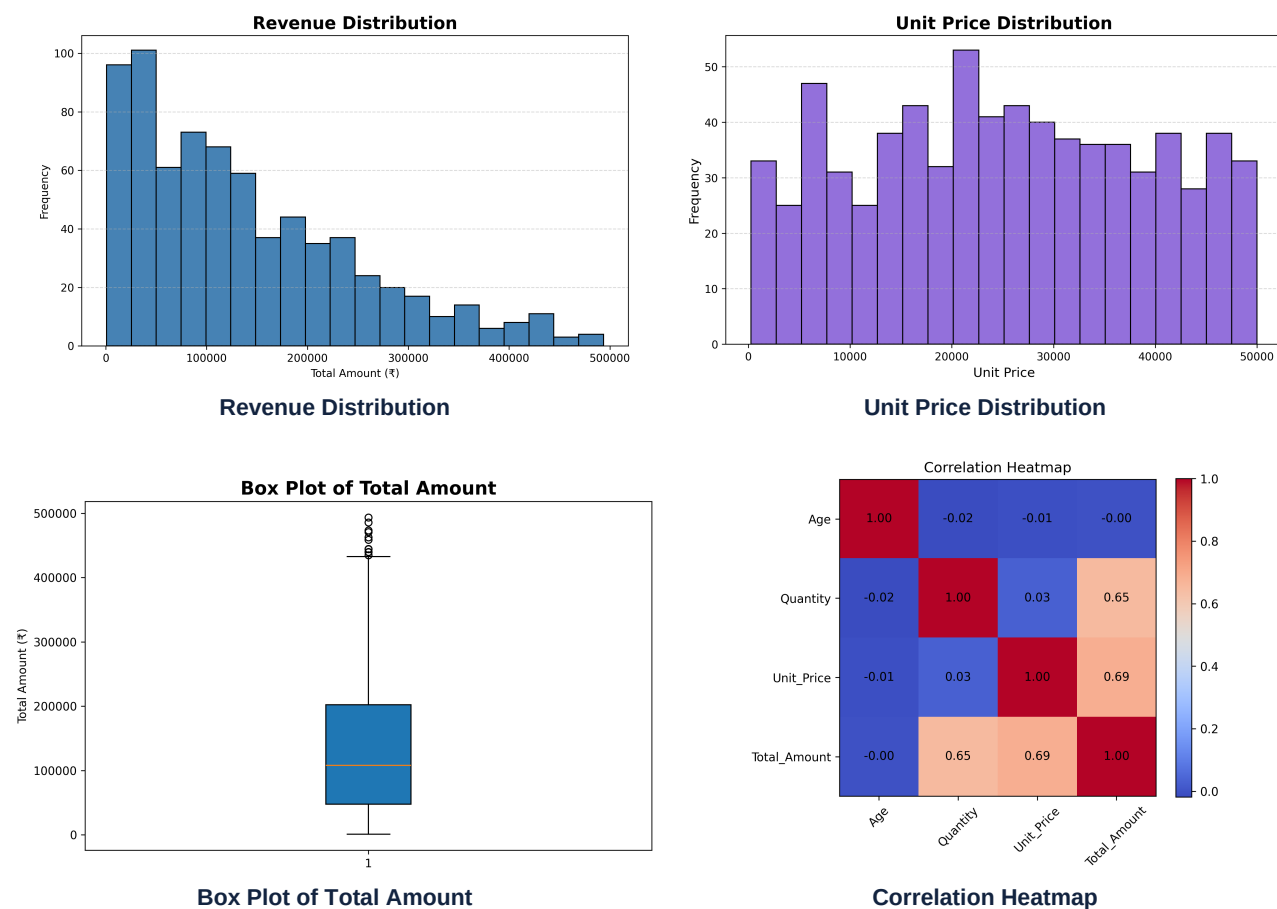


Monthly Quantity Sold

Revenue and orders dip sharply in September before rebounding to a strong October–November peak. Quantity sold trends upward toward year-end, pointing to a Q4 seasonal surge.

4. Revenue Distribution & Correlation

Transaction-level patterns and relationships between key variables.



INTERPRETATION

- Most transactions fall under ₹50,000, with a long tail of high-value orders up to ₹500,000.
- Quantity ($r = 0.65$) and Unit Price ($r = 0.69$) both show a strong positive correlation with Total Amount.
- Age shows almost no correlation ($r \approx 0.00$) with spend — purchasing behaviour is age-independent.
- Box plot outliers above ₹430,000 represent high-value transactions worth targeting for loyalty offers.

5. Conclusion

This EDA examined the E-Commerce Sales dataset across customer demographics, product performance, city-wise sales, monthly trends, and revenue distribution. Quantity purchased and unit price emerged as the primary drivers of revenue, while customer age had negligible influence. These findings, detailed further in the accompanying Business Insights report, demonstrate practical, portfolio-ready data analysis skills using Python, Pandas, NumPy and Matplotlib.

Tools: Python, Google Colab, Pandas, NumPy, Matplotlib, Microsoft Word, Git & GitHub

Skills: Data Cleaning, Data Wrangling, EDA, Data Visualization, Statistical Analysis, Correlation Analysis, Business Insight Generation, Report Writing, GitHub Project Management